Anirban Chakraborty

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Statement of Purpose

I am a postdoctoral scholar in the Department of Public Health Sciences at the Medical University of South Carolina, working with Dr. Souvik Seal. I develop statistical methods for single-cell and spatial omics, with a focus on biomarker discovery and replicate-aware multi-sample analysis, building on a background in scalable Bayesian methods for spatial data.

Research interest and Expertise

Statistical methodology
Gaussian and non-Gaussian processes
Bayesian transport maps
Variable selection
Scalable computations
Variational inference

Data applications
Single cell and spatial omics
– Biomarker discovery
– Molecular colocalization
– Multi-sample analysis
Environmental modeling
– Geophysical model emulation and calibration
– Atmospheric filtering

Education

Texas A&M University, College Station, Texas, USA.
Ph.D., Statistics. Advisor- Dr. Matthias Katzfuss. August 2019 - June 2024
- Thesis. Scalable Statistical Methods for Large Spatial and Spatio-Temporal Data.

Indian Statistical Institute, Kolkata, India.
Master of Statistics (M. Stat). July 2016 - June 2018.

University of Calcutta, Kolkata, India.
Bachelor of Science (B.Sc.). July 2013 - May 2016.
- Major in Statistics, minor in Mathematics & Economics.

Work experience

Medical University of South Carolina, Charleston, SC, USA
Postdoctoral scholar. Supervisor - Dr. Souvik Seal. July 2025 - present.
- I am currently developing scalable biomarker identification methods for single cell and spatial omics data using advanced statistical techniques, including kernel machine regression, spike and slab prior distribution, spatially aware dimension reduction.

The University of Texas MD Anderson Cancer Center, Houston, TX, USA
Postdoctoral scholar. Supervisor - Dr. Wenyi Wang. August 2024 - July 2025
- I contributed to benchmarking and developments of transcriptomic analysis software in Dr. Wang's lab.

Texas A&M University, College Station, Texas, USA.
Graduate Assistant & Lecturer, Department of Statistics. August 2019 - August 2024
- I served as a graduate instructor as well as graduate teaching assistant.
- As a research assistant funded by Jet Propulsion Lab (JPL), I collaborated with scientists in JPL to develop nonlinear uncertainty quantification for OCO-2/3 satellite.

GlaxoSmithKline Pvt. Ltd., Bengaluru, India.

Associate statistician. Therapeutic area: Oncology. July 2018 - July 2019.
- I developed analysis plans, performed quality control for 2 Oncology projects.
- I performed simulation and analysis scenario generation of a respiratory disease-based clinical trial in collaboration with my colleagues.

Eli Lilly and Company, Indianapolis, Indiana, USA. May 2023 - August 2023.
Ph.D. Statistics Intern, Therapeutic area: Diabetes.
- Developed a novel Bayesian non-parametric longitudinal dose response model. This method showed superior performance over existing methods, e.g., EMAX, ITP, tITP.
- Validated our results on a Phase III Diabetes study.

Teaching experience

Graduate Lecturer, Texas A&M University. Fall 2023
- Taught STAT 302 (Section 506), with an enrollment of 43 students. Topics included probability, estimation, hypothesis testing, and categorical data analysis.
- Collaborated with instructors across multiple sections to develop common midterm examinations.

Graduate Teaching Assistant, Texas A&M University. Fall 2019 – Summer 2024
- Supported instruction in undergraduate and graduate statistics courses.
- Prepared reproducible homework solutions for STAT 611 (Statistical Inference).
- Assisted instructors with the transition to online instruction and adapted grading and discussion activities for online delivery, beginning during the COVID-19 pandemic.

Grant application

Chakraborty A. (Principal Investigator), Seal S., Angel P., *Scalable statistical dimension reduction and biomarker discovery methods for multi slice spatial omics analysis.*, 2026. (Decision pending)

Accepted publications/ under revision

- Asterisk ($\star$) denotes equal contribution.
- Co-authors of independent collaborations are underlined.

[12] **Chakraborty A.**, Mattila C., Ghosh D., Neelon B., Seal S. CBKMR: A Copula-based Bayesian Kernel Machine Regression Framework for Optimal Marker Detection in Omics Data. *Under revision at Annals of Applied Statistics.* 2026. [Link]

[11] Chakraborty A., Mandal A., **Chakraborty A.** Probabilistically-routed Bayesian Additive Spanning Trees for Learning on Constrained Domains. *Forty-third International Conference on Machine Learning (ICML)*, 2026. [Link]

[10] Chakraborty A., **Chakraborty A.** Hierarchical Transit Compartment Model to Describe Absorption Delay in Orally Administered Drugs in Heterogeneous Populations. *Therapeutic Innovation & Regulatory Science*, 2026.

[9] **Chakraborty A.***, Chakraborty A.*. Efficient and Scalable Tobit Gaussian Process Regression for Modeling Air Quality Data. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026. [Link]

[8] **Chakraborty A.**, Katzfuss M. Learning non-Gaussian spatial distributions via Bayesian transport maps with parametric shrinkage. *Journal of Agricultural, Biological and Environmental Statistics (JABES)*, 2025. [Link]

[7] Chakraborty A., **Chakraborty A.** Meta-Learning with Adaptive Borrowing for High Dimensional Moment Condition Model. *2025 IEEE International Conference on Big Data (BigData)*, 2025. [Link]

[6] **Chakraborty A.***, Chakraborty A.*. Scalable model-based gaussian process clustering. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024. [Link]

[5] **Chakraborty A.**, Katzfuss M. High-dimensional nonlinear spatio-temporal filtering by compressing hierarchical sparse Cholesky factors. *Journal of Data Science*, 2022. [Link]

[4] **Chakraborty A.**, Katzfuss M., Guinness J. Ordered conditional approximation of Potts models. *Spatial Statistics*, 2022. [Link]

Submitted manuscripts

[3] Mattila C.*, **Chakraborty A.***, Angel P., Cao S., Sonnawane K., Hill E., Chung D., Neelon B., Seal S. FusedFCR: A Fused Forward Continuation-Ratio model for marker selection along cell-fate trajectories. *Submitted to Genome Research*, 2026. [Link]

[2] **Chakraborty A.**, Neelon B., Lawson A., Angel P., Chung D., Seal S. MSGPCA: Multi-Slice Graph PCA for replicate-aware Spatial Omics analysis. *Submitted to bioRxiv*, 2026. [Link]

[1] Seal S., **Chakraborty A.**, Mattila C., Rubinstein M., Angel P., Ghosh D., Chung D., Neelon B. RKMR: A Rapid Kernel Machine Regression Framework for Optimal Marker Detection in Spatial Omics Data. *Submitted to bioarxiv*, 2026. [Link]

Posters and Presentations

MSGPCA: Multi-Slice Graph PCA for replicate-aware Spatial Omics analysis. AI in Clinical and Translational Research, 2026.

CBKMR: A Copula-based Bayesian Kernel Machine Regression Framework for Optimal Marker Detection in Omics Data. STATGEN, 2026.

DeMixT 2.0: R package for semi-reference-based transcriptomic deconvolution of heterogeneous tumor samples. MD Anderson Cancer Center Support Grant retreat, 2025.

Learning Non-Gaussian Spatial Distributions Via Bayesian Transport Maps with Parametric Shrinkage. ENVR workshop, 2024.

Bayesian nonparametric calibration of climate models. Spatial Statistics workshop, July 2023.

Ordered conditional approximation for Potts models. Joint Statistical Meetings, 2022.

Awards

• **External Travel Awards.** Statistical Genetics workshop (STATGEN, 2026); Spatial Statistics workshop (2023); Envrionmental Statistics workshop (ENVR, 2022, 2024).

• **Texas A&M University.** Research and Presentation Travel Award, School of Graduate studies (2024); Strategic University Partnership Program (SURP) by Jet Propulsion Laboratory (2021-2023); Lechner Competitive Scholarship (2019).

• **Government of India.** Merit-based Inspire Scholarship for Higher Education, Department of Science & Technology, for being in the top 1% in the country in the higher secondary examination (2013-16).

• **Other scholastic achievements.** Academic fellowship, Indian Statistical Institute (2016-2018); All India Rank 10 in Joint Admission Test, Mathematical Statistics, Indian Institute of Technology (2016).

Softwares

[R] FusedFCR [Link], CBKMR [Link], Compressed hierarchical Vecchia filter [Link]

[Python] Parametric shrinkage transport map [Link]

Service to Profession

• **Peer reviewer.** Annals of Applied Statistics; Statistical Analysis and Data Mining;

IEEE International Conference on Acoustics, Speech and Signal Processing, Nature Scientific Reports; Journal of Clinical Oncology.

• **Session Organizer.** "Recent advances in uncertainty quantification for complex systems" (Joint Statistical Meetings, August, 2023).

Reference

Matthias Katzfuss,
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Professor,
Department of Statistics,
University of Wisconsin–Madison (previously at Texas A&M University),
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